随机偏微分方程测试.

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一. (i) $\tau:\Omega\to[0,+\infty)$ 是 $(\Omega,(\mathcal{F}_t)_{t\geq 0},\mathbb{P})$ 的停时. 若 $\forall t\geq 0$:

$$\{\omega\in\Omega \mid \tau(\omega)\leq t\}\in\mathcal{F}_t$$

与其适定的域 $\mathcal{F}_\tau$ 如下定义:

$$\mathcal{F}_\tau=\{A\subseteq\Omega,\ \forall t\geq 0,\ A\cap\{\tau\leq t\}\in\mathcal{F}_t\}.$$

(ii) $\mathcal{W}:\{\omega\in C(\mathbb{R}_+,\mathbb{R}^d)\}$ 是概率空间. 转移算子 $\theta_t:\mathcal{W}\to\mathcal{W}$ $(t\geq 0)$

: $\theta_t(\omega)(s)=\omega(t+s)$; $\mathcal{F}_t$ · 右垂直滤子.

称. 一簇 $\mathcal{W}$ 上概率 $\{\mathbb{P}_x\}_{x\in\mathbb{R}^d}$ 有马氏性 (Markov Property), 若 $\forall x\in\mathbb{R}^d$, $t\geq 0$ 任一与 $(\mathcal{W},(\mathcal{F}_t)_{t\geq 0})$ 适定的过程 Y, 成立:

$$\mathbb{E}_x(Y\circ\theta_t\mid\mathcal{F}_t)=\mathbb{E}_{X_t}(Y)$$

其中. $\mathbb{E}_x$ 表示 $\mathbb{P}_x$ 概率下期望. , $\mathbb{E}_{X_t}(Y)(\omega):=\mathbb{E}_{\omega(t)}(Y)$

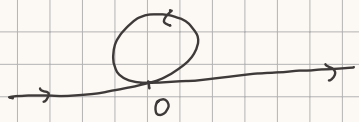
称. $\{\mathbb{P}_x\}_{x\in\mathbb{R}^d}$ 有强马氏性. (Strong Markov Property) . 若 $\forall x\in\mathbb{R}^d$, 任一与 $(\mathcal{W},(\mathcal{F}_t)_{t\geq 0})$ 适定的过程 Y, 有界停时. τ. 成立.

$$\mathbb{E}_x(Y\circ\theta_\tau\mid\mathcal{F}_\tau)=\mathbb{E}_{X_\tau}(Y)$$

其中: $\theta_\tau(\omega)(t)=\omega(t+\tau(\omega))$

马氏性代表一种无记忆性。

(iii) 设: $I:\mathbb{R}\to\mathbb{R}^2$:



I 在 $\mathbb{R}^2$ 有自交点 O

考虑 $X_t=I(W_t)$, W_t 是 $\mathbb{R}$ 上的 B.M.

$\forall x\in\operatorname{im} I\setminus\{0\}$, $\mathbb{P}_x=$ law of W_t starting at $I^{-1}(x)$

$\forall x=0$, $I^{-1}(0)$ 有两个点 (A_1, A_2). $\mathbb{P}_0 = \frac{1}{2}$ law of W_t -- A_1

$+\frac{1}{2}$ law. W_t -- A_2

$\forall s>0$

$$\mathbb{E}_x(f(X_t)\mid \mathcal{F}_s) = \operatorname*{Avg}_{y\in I^{-1}(x)} \mathbb{E}_y(f\circ I(W_t)\mid \mathcal{F}_s)$$

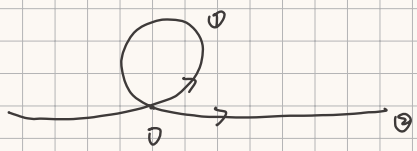
$$\underset{(W_t\text{ 的马氏性})}{=} \operatorname*{Avg}_{y\in I^{-1}(x)} \mathbb{E}_{W_s}(f\circ I(W_{t-s}))$$

$$= \mathbb{E}_{X_s} f(X_{t-s})$$

故其有马氏性

但取 $\tau = \inf\{t,\ X_t = 0\}$

则. $\mathbb{E}(\cdots \mid \mathcal{F}_{\le\tau}) \ne \mathbb{E}(\quad \mid X_\tau)$



这是由于. $\mathbb{E}(\cdots \mid X_\tau)$ 在 0 点后有 $\frac{1}{2}$ 概率 $I^{-1}(0) = A_1$

有 $\frac{1}{2}$ 概率 $I^{-1}(0) = A_2$.

而 $\mathbb{E}(\cdots \mid \mathcal{F}_{\le\tau})$ 可得: $I^{-1}(0)$.

故其没有强马式性.

二. (1) $(W_t)_{t\geq 0}$ 满足 ① $0\leq s<t$, W_t-W_s 分布为 $N(0, t-s)$

即方差为 $t-s$ 的高斯分布; 且 W_t-W_s 独立于 $(\mathcal{F}_\theta)_{0\leq\theta\leq u}$

② a.e. $w\in\Omega$, $W_t(w)$ 关于 t 连续.

(2) $\forall s<t$,

$$\mathbb{E}(W_t|\mathcal{F}_s) = \mathbb{E}(W_t-W_s|\mathcal{F}_s) + W_s$$

由于 $W_t-W_s \perp \mathcal{F}_s$, $\mathbb{E}(W_t-W_s|\mathcal{F}_s) = \mathbb{E}(W_t-W_s) = 0$

$\Rightarrow \mathbb{E}(W_t|\mathcal{F}_s) = W_s$. 故 W_t 为 $\mathcal{F}_t$-Martingale

(3) $\langle B_t, B_t\rangle = \langle W_{t+T}-W_T, W_{t+T}-W_T\rangle$

由于 $\langle W_t, W_s\rangle = t\wedge s$

计算得. $\langle B_t, B_t\rangle = t+T-2T+T = t$

且由 W_t 的连续性知 B_t 的连续性.

由 levy thm. (B_t) 为 $\mathcal{G}_t$-Brownian Motion.

由. W_t (B.M.) 的强马氏性.

$$\mathbb{E}(W_{t+T}-W_T|\mathcal{F}_T) = \mathbb{E}_{W_T}(W_{t+T}-W_T) = \mathbb{E}(W_t) = 0$$

即 $\mathbb{E}(B_t|\mathcal{G}_0) = 0$, 故 (B_t) 与 $\mathcal{G}_0$ 独立.

三. (1) $e^{\lambda|x|}$ 是下凸函数. 由Jensen不等式:

$$\mathbb{E}(e^{\lambda|W_t|}|\mathcal{F}_s) \geq e^{\lambda|\mathbb{E}(W_t|\mathcal{F}_s)|} = e^{\lambda|W_s|}, \quad \forall t>s$$

故 $\mathbb{E}(X_t|\mathcal{F}_s) \geq X_s$, $\forall t>s$. X_t 为下鞅.

(2)

由Doob不等式, 结合(1) $e^{\lambda|W_s|}$ 为下鞅:

$$e^{\lambda x}\cdot \mathbb{P}(\sup_{0\leq s\leq t}|W_s|>x) \leq \mathbb{E}(e^{\lambda|W_s|})$$

$$= \frac{1}{\sqrt{2\pi t}}\int_{-\infty}^{+\infty} e^{\lambda|z|-\frac{z^2}{2t}}dz = \frac{2}{\sqrt{2\pi t}}\int_0^{\infty} e^{\frac{1}{2t}(z-\lambda t)^2+\frac{\lambda^2 t}{2}}dz$$

$$\leq 2e^{\frac{\lambda^2 t}{2}}\int_{-\infty}^{+\infty} e^{-\frac{z^2}{2t}}\Big/\sqrt{2\pi t}\,dz \leq 2e^{\frac{\lambda^2 t}{2}}$$

故 $\mathbb{P}(\sup_{0\leq s\leq t}|W_s|>x) \leq 2e^{\frac{\lambda^2 t}{2}-\lambda x}$

取 $\lambda=\frac{x}{t}>0$, 即 $\mathbb{P}(\sup_{0\leq s\leq t}|W_s|>x) \leq 2e^{-\frac{x^2}{2t}}$

(3) 记 $T(t)=\langle X,X\rangle_t = \int_0^t |b_s|^2 ds \geq \frac{t}{4}$ $(|b|\geq\frac{1}{2})$

故连续鞅 $X_t = W_{T(t)}$, 由(2):

$$\mathbb{P}(\sup_{0\leq s\leq t}|X_s|>x) \leq \mathbb{P}(\sup_{0\leq s\leq T(t)}|W_s|>x)$$

$$\le \mathbb{P}\left(\sup_{0\le s\le \frac{t}{4}} |W_s| > x\right) \le 2e^{-\frac{2x^2}{t}}.$$

(4) $\mathbb{P}(\tau_1 > t) = \mathbb{P}\left(\sup_{0\le s\le t} |W_s| < 1\right)$

$\forall n\in\mathbb{N}$. 若: $|W_s| < 1$. $\forall 0\le s\le t$,

取 $t_0=0$, $t_i = \frac{it}{n}$ $(1\le i\le n)$, 故 $|W_{t_{i+1}} - W_{t_i}| < 2$

$\Rightarrow \mathbb{P}(\tau_1 > t) \le \mathbb{P}\left(|W_{t_i} - W_{t_{i-1}}| < 2,\ 1\le i\le n\right)$

由B.M.独立性

$$\le \prod_{i=1}^{n} \mathbb{P}(|W_{t_i} - W_{t_{i-1}}| < 2) = \mathbb{P}\left(|W_{\frac{t}{n}}| < 2\right)^n$$

而 $\mathbb{P}\left(|W_{\frac{t}{n}}| < 2\right) = \sqrt{\frac{n}{2\pi t}} \int 1_{\{|x|<2\}} e^{-\frac{nx^2}{2t}} dx$

$$\le \sqrt{\frac{n}{2\pi t}}$$

故 $\mathbb{P}(\tau_1 > t) \le \left(\sqrt{\frac{n}{2\pi t}}\right)^n \quad \forall n\in\mathbb{N}$

特别的. $\mathbb{P}(\tau_1 > t) \le C t^{-2}$.

$\mathbb{E}(\tau_1) = \int_0^\infty \mathbb{P}(\tau_1 > t)\, dt \le 1 + \int_1^{+\infty} C t^{-2} dt \le C$

下归纳证明: $\mathbb{E}(\tau_1^n) \le n!\, C^n$

$n \Rightarrow n+1$

$\mathbb{E}(\tau_1^{n+1}) = (n+1)!\, \mathbb{E}\left(\int 1_{\{0<t_1<\cdots<t_{n+1}<\tau_1\}} dt_1 dt_2 \cdots dt_{n+1}\right)$

$$= (n+1)\cdot n!\ \mathbb{E}\left(\int \mathbb{1}_{\{0<t_1<\cdots<t_n<\tau_R\}}\cdot \mathbb{1}_{\{t_n<t_{n+1}<\tau_1\}}\, dt_1\cdots dt_{n+1}\right)$$

$$= (n+1)\cdot n! \int \mathbb{E}\left(\mathbb{1}_{\{0<t_1<\cdots<t_n<\tau_R\}}\cdot \mathbb{E}\left(\mathbb{1}_{\{t_n<t_{n+1}<\tau_1\}}\,|\,\mathcal{F}_{t_n}\right)\right) dt_1\cdots dt_{n+1}$$

由马氏性: $\int \mathbb{E}\left(\mathbb{1}_{\{t_n<t_{n+1}<\tau_1\}}\,|\,\mathcal{F}_{t_n}\right) dt_{n+1} = \mathbb{E}\left((\tau_1-t_n)^+\,|\,\mathcal{F}_{t_n}\right)$

$$= \mathbb{E}_{X_{t_n}}(\tau_1) \le \mathbb{E}(\tau_2) \le C.$$

由归纳假设: $n!\int \mathbb{E}\left(\mathbb{1}_{\{0<t_1<\cdots<t_n<\tau_R\}}\right) dt_1\cdots dt_n = \mathbb{E}(\tau_1^n) \le n!\cdot C^n$

故 $\mathbb{E}(\tau_1^{n+1}) \le (n+1)!\, C^{n+1}$

取 $\mu \le C^{-1}$. 故 $\mathbb{E}(\mu^{n+1}\tau_1^{n+1}) \le (n+1)!\,(\mu C)^{n+1}$

$$\mathbb{E}(e^{\mu\tau_1}) \le \sum_n (\mu C)^n \le \frac{1}{1-\mu C} < +\infty$$

(5) $(W_t^i)_{1\le i\le d}$ 是 d-维 B.M.

$\tau_1 = \inf\{t \,|\, \|(W_t^i)\|_{\mathbb{R}^d} > 1\}$, 记 $\tau_1' = \inf\{t, |W_t^1| > 1\}$

由于 $|W_t^1| \le \left(\sum |W_t^i|^2\right)^{\frac{1}{2}} = \|(W_t^i)\|_{\mathbb{R}^d}$

故 $\tau_1 \le \tau_1'$

W_t^1 一为一维 B.M., 由(4)知

$$\mathbb{E}\, e^{\mu\tau_1} \le \mathbb{E}\, e^{\mu\tau_1'} < +\infty$$

故 d-维有相应结论.

四. 记 $dQ^1 = \exp\left(-\int_0^\infty \dot{h}_s dW_s - \frac{1}{2}\int_0^\infty \dot{h}_s^2 ds\right) d\mathbb{P}$

由 Girsonov thm. $W_t + h_t$ 是 Q^1-概率下的 B.M.

故 $\mathbb{E}_{Q^1}(\phi(W_t+h_t)) = \mathbb{E}_{\mathbb{P}}(\phi(W_t)) = \mathbb{E}_Q(\phi(W_t+h_t))$

故 $Q = Q^1$

$$H(Q|\mathbb{P}) = \mathbb{E}_Q\left(-\int_0^\infty \dot{h}_s dW_s - \frac{1}{2}\int_0^\infty \dot{h}_s^2 ds\right).$$

$$= \mathbb{E}_Q\left(-\int_0^\infty \dot{h}_s d(W_s+h_s)\right) + \frac{1}{2}\int_0^\infty \dot{h}_s^2 ds$$

由于 W_s+h_s 在 Q-下为 Martingale，故.

$$H(Q|\mathbb{P}) = \frac{1}{2}\int_0^\infty \dot{h}_s^2 ds$$

五：最五可由题六推出.

事实上. 由六，取 $D=\{|x|<1\}$，$L=\Delta$ $(a_{ij})=I$

若 $u \in C^2(D)\cap C(\bar{D})$，$\Delta u=0$，且 $u|_{\partial D}=0$，$u(0)=1$.

故 $\sup_D u \geq 1 > \sup_{\partial D} u = 0$，最大值不在边界取到.

$\Rightarrow$ u 在 D 上为常数. 矛盾

六：设 $(a_{ij}) = ({}^t\sigma)(\sigma)$，考虑 SDE.

$$\begin{cases} dX_t^i = \sigma_{ij}(X_t)\,dW_t^j + \frac{1}{2}b_i(X_t)\,dt. \\ t=0: \quad X_0 \equiv x_0 \end{cases}$$

由 Ito-formula:

$$u(X_t) - u(X_0) = \int_0^t \nabla u(\vec{X}_s)\cdot d\vec{X}_s + \frac{1}{2}\int_0^t u_{ij}(\vec{X}_s)\,d\langle \vec{X}_s, \vec{X}_s\rangle$$

$$= \int_0^t \nabla u\cdot(\sigma)\,dW_s + \frac{1}{2}\int_0^t b_i(X_s)\,\partial_i u(X_s)\,ds + \frac{1}{2}\int a_{ij}\cdot u_{ij}(X_s)\,ds$$

$$Lu = \sum a_{ij}u_{ij} + \sum b_i u_i = 0$$

停时 $\tau = \inf\{t \mid X_t \notin D\}$.

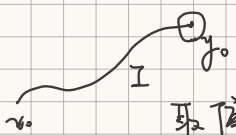
故 $u(X_t) - u(X_0) = \int_0^t \nabla u\cdot(\sigma)\,dW_s \qquad t \le \tau$.

若 $\exists\, x_0 \in \operatorname{int} D$. $u(x_0) = \sup u$

对 $\forall\, y_0 \in \operatorname{int} D$, $u(y_0) < u(x_0) = \sup u$

$u \in C(\bar{D})$. 故 $\exists\, \epsilon > 0$. s.t $\forall |y - y_0| \le \epsilon$. $|u(x_0) - u(y)| \le \frac{1}{2}|u(x_0) - u(y_0)|$

取连接道路 $I \subseteq \operatorname{int} D$, $I_0 = x_0$. $I_1 = y_0$, 不妨设 ϵ 充小, $N_\epsilon(I) \subseteq \operatorname{int} D$.



由 support 定理: $\mathbb{P}\left(\sup_{t\in[0,1]} |X_t - I_t| \le \epsilon\right) > 0 \quad (*)$

取停时 $T = \inf\{t \le \tau \mid X_t \in B_\epsilon(y_0)\}$

由 $(*)$ 知 $\mathbb{P}(t < T) > 0$

由鞅性质：$\mathbb{E}(u(X_T)-u(x_0)) = \mathbb{E}\left(\int_0^T \nabla u(B_s)\,dW_s\right) = 0$.

但 $\mathbb{E}(u(X_T)-u(x_0)) = \mathbb{E}(\mathbb{1}_{\{t<T\}}(u(X_T)-u(x_0))$ ··· ①

$+\mathbb{E}(\mathbb{1}_{\{t=T\}}(u(X_T)-u(x_0))$ ··· ②

② ≤ 0 ，① $\le -\mathbb{P}(\mathbb{1}_{\{t<T\}})\cdot\frac{1}{2}\cdot|u(x_0)-y_0| < 0$ 矛盾！

故 $\forall y\in \mathrm{int}\,D$. $u(y)=u(x_0)$

$\Rightarrow u\equiv \mathrm{const}$ on D

若 u 不为常值，则其最大值只能在边界上取到

七：坐标变换：$t\to T-t$.

只需证明：$u(t,x)=\inf_{\alpha\in\mathcal{A}} \mathbb{E}_x g(X_t^\alpha)$ 是方程的解：

$$\begin{cases} \partial_t u - \frac{1}{2}\inf_{\alpha\in A}\{[\sigma\sigma^T(\alpha)]_{ij}\partial_{ij}u\} = 0 \\ u(0)=g \end{cases}$$

首先说明. $\forall\, 0<s<t$, $u(t,x)=\inf_{\alpha\in\mathcal{A}} \mathbb{E}_x u(t-s, X_s^\alpha)$

记：$u^\alpha(t,x)=\mathbb{E}_x g(X_t^\alpha)$

由 X_t^α 的强马氏性：$\mathbb{E}_x(g(X_t^\alpha)|\mathcal{F}_s) = \mathbb{E}_{X_s^\alpha}(g(X_{t-s}^{\alpha\cdot\theta_s})) = u^{\alpha\cdot\theta_s}(t-s, X_s^\alpha)$

故 $u^\alpha(t,x)=\mathbb{E}_x(g(X_t^\alpha)) = \mathbb{E}_x(\mathbb{E}_{X_s^\alpha}(g(X_{t-s}^{\alpha\cdot\theta_s})) = \mathbb{E}_x(u^{\alpha\cdot\theta_s}(t-s, X_s^\alpha))$

$u(x,t)=\inf_{\alpha\in\mathcal{A}} u^\alpha(t,x)$ 故 $u(x,t)=\inf_{\alpha\in\mathcal{A}} \mathbb{E}_x(u^{\alpha\cdot\theta_s}(t-s, X_s^\alpha))$

若 $\alpha|_{[0,s]}=\beta|_{[0,s]}$ 则. $X_\theta^\alpha = X_\theta^\beta$, $\forall\theta\in[0,s]$ （解的唯一性. 或 $\theta\in C(\mathbb{R}^d)$）

$$\Rightarrow u(x,t) = \inf_{\substack{\beta:[0,\delta]\to A \\ \gamma:[0,t-s]\to A}} \mathbb{E}_x\left(u^{\gamma}(t-s, X_\delta^{\beta})\right)$$

[移：α 拆分为：$\beta = \alpha\big|_{[0,s]}$，$\gamma = \alpha\circ\theta_s\big|_{[0,t-s]} = \alpha\big|_{[s,t]}$]

$$= \inf_{\beta:[0,\delta]\to A} \mathbb{E}_x\left(u(t-s, X_\delta^{\beta})\ \left[\inf_{\gamma} u^{\gamma}(t-s,x) = u(t-s,x)\right]\right)$$

$\forall t$, $\quad u(t+s,x) = \inf_{\alpha\in A} \mathbb{E}_x u(t, X_s^{\alpha})$

由 Ito-Formula：$u(t, X_s^{\alpha}) - u(t, X_0^{\alpha}) = \int_0^s \nabla u\cdot\sigma\, dW_\theta + \frac{1}{2}\int_0^s {}^t\sigma\sigma : \partial_{ij}u\, d\theta$

故 $\mathbb{E}_x u(t, X_s^{\alpha}) - u(t,x) = \frac{1}{2}\cdot\int_0^s \mathbb{E}_x\left({}^t\sigma\sigma : \partial_{ij}u(X_\theta^{\alpha})\right) d\theta$

再 $\inf_{\alpha\in A}$ $\Rightarrow$ $u(t+s,x) - u(t,x) = \frac{1}{2}\inf_{\alpha\in A}\int_0^s \mathbb{E}_x({}^t\sigma\sigma : \partial_{ij}u)(X_\theta^{\alpha})\, d\theta$

故 $\partial_t u(t,x) = \lim_{s\to 0}\frac{u(t+s,x) - u(t,x)}{s} = \frac{1}{2}\cdot\lim_{s\to 0}\frac{1}{s}\inf_{\alpha\in A}\int_0^s \mathbb{E}_x({}^t\sigma\sigma : \partial_{ij}u)(X_\theta^{\alpha})\, d\theta$

由连续性：$|\theta|\le s\to 0$，$\forall\alpha$，$\mathbb{E}_x({}^t\sigma\sigma : \partial_{ij}u)(X_0^{\alpha}) \to {}^t\sigma\sigma^{\alpha} : \partial_{ij}u(x)$

故 $\partial_t u(t,x) = \frac{1}{2}\inf_{\alpha\in A} {}^t\sigma\sigma^{(\alpha)} : \partial_{ij}u(x)$ 满足方程.